

Insurance Claim Fraud Prediction Using Machine Learning

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1. Overview of Problem Statement

Insurance fraud is a major issue in the insurance industry and can lead to significant financial losses. Detecting fraudulent insurance claims manually is difficult because of the large amount of claim data and complex fraud patterns. Therefore, machine learning techniques can be used to identify fraudulent insurance claims accurately and efficiently.

2. Objective

To develop a machine learning model that predicts whether an insurance claim is fraudulent or non-fraudulent using insurance claim data.

3. Data Description

Source

Kaggle Dataset-Insurance Claims - Fraud Detection

Features

The dataset contains customer details, vehicle details, incident information, claim amount details, and fraud report status.

Some important features include:

- age
- policy_state
- policy_csl
- policy_deductable
- insured_sex
- insured_education_level
- insured_occupation
- insured_hobbies
- incident_type
- collision_type
- incident_severity
- authorities_contacted

- total_claim_amount
- injury_claim
- property_claim
- vehicle_claim
- fraud_reported

4. Data Collection

The dataset was imported from Kaggle using a CSV file.

Python libraries such as Pandas and NumPy were used to load and analyze the dataset.

Initial analysis was performed to understand:

- Data shape
- Data types
- Missing values
- Distribution of features

5. Data Preprocessing – Data Cleaning

The following preprocessing steps were performed:

- Replaced '?' values with NaN
- Handled missing values using mode imputation
- Removed unnecessary columns such as policy_number and incident_location
- Checked duplicate values
- Detected and removed outliers using IQR method
- Encoded categorical features using Label Encoding

6. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships in the dataset.

Visualizations Used

- Histogram
- Boxplot
- Heatmap Correlation
- Count Plot
- Bar Plot
- KDE Plot

Insights

- Fraud cases were fewer compared to non-fraud cases.
- Certain incident types had higher fraud occurrence.
- Claim amount distribution showed the presence of outliers.
- Correlation heatmap helped identify relationships between features.

7. Feature Engineering

Categorical features were converted into numerical values using Label Encoding.

Examples:

- Male/Female → numerical values
- YES/NO → numerical values

This conversion helped machine learning models process the data efficiently.

8. Feature Selection

Advanced feature selection techniques were not applied in this project.

However, unnecessary columns were removed during preprocessing to improve model efficiency.

9. Split Data into Training and Testing Sets

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This helped evaluate the model performance on unseen data.

10. Feature Scaling

Feature scaling was not applied because the selected models such as Decision Tree and Random Forest do not strictly require scaling.

11. Build the ML Model

The following machine learning classification algorithms were implemented:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. Naive Bayes Classifier

These models were trained using the training dataset.

12. Model Evaluation

The models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Precision
- Recall
- F1-Score

Model Results

- Logistic Regression Accuracy: 77%
- Decision Tree Accuracy: 78%
- Random Forest Accuracy: 75.5%
- Naive Bayes Accuracy: 69%

Decision Tree Classifier achieved the highest accuracy for this dataset.

13. Hyperparameter Tuning and Pipeline

Hyperparameter tuning and pipelines were not implemented in this mini-project because the focus was on basic machine learning algorithms covered in the syllabus.

14. Save the Model

The trained model was saved using Joblib.

Saved file:

- insurance_fraud_model.pkl

This allows the trained model to be reused later without retraining.

15. Test with Unseen Data

The trained model was tested using testing data that was not used during training.

This helped evaluate how well the model performs on unseen insurance claim records.

16. Interpretation of Results (Conclusion)

This project successfully implemented machine learning techniques for insurance fraud prediction.

Different classification algorithms were trained and evaluated. Among all models, Decision Tree Classifier achieved the best performance for this dataset.

The project demonstrated how machine learning can help automate fraud detection and reduce manual effort in the insurance industry.

Limitations

- Dataset size was limited.
- Fraud cases were imbalanced.
- Advanced optimization techniques were not used.

17. Future Work

The project can be improved in the future by:

- Using advanced machine learning and deep learning models
- Applying hyperparameter tuning
- Handling imbalanced datasets using resampling techniques
- Adding more real-world insurance data
- Building a web application for live fraud prediction

Project Type

- Machine Learning Project
- Supervised Learning
- Binary Classification